Qiaoyuan **Zheng**

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qiaoyuan-zheng.com | Zurich, Switzerland | MSc in Data Science



MSc student in Data Science at ETH Zürich, researching reliable LLM evaluation, interpretable capability assessment, and privacy in language agents.

Education

ETH Zürich — *Zurich, Switzerland*
M.Sc. Data Science — *2025 - Present*

- GPA:5/6
- Course: Probabilistic AI | Large Language Models | Big Data | Computational Statistics | Large-Scale AI Engineering
- Special Project-oriented course: AI Center Project | Research in DS | Data Science Lab

Sorbonne University — *Paris, France*
B.Sc. Computer Science/Mathematics — *2022 - 2025*

- Math Course: Linear Algebra | Digital and Functional Series | Topology | Theory of Measurement | Theory of Probability
- CS Course: Algorithms | Data Structure | Object-oriented Programming | Functional Programming | Advanced C
- GPA: 17.10/20, 3.96/4.0 (WES), L3 Mention:Bien

Research Experience

ETH AI Center — *Zurich, Switzerland*
Research Assistant — *2026 - Present*

- Develop a multilingual benchmark for evaluating legal compliance in large language models
- Building reproducible pipelines for scenario generation, independent review, and model evaluation
- Collaborate with legal experts to assess scenario plausibility and legal validity
- Investigate benchmark applications for LLM post-training

Apertus | LLM Evaluation | Python

ESIEE Paris — *Paris, France*
Research Assistant · Github — *2023 - 2023*

- Implemented Transformer-based models and PatchTST for time-series forecasting under the supervision of Dr. You Jiang
- Adapted data and training pipelines, comparing model performance and computational efficiency against U-Net 24 on synthetic datasets

Python | PyTorch | Time-Series Forecasting

Publications

POLAR-Bench: A Diagnostic Benchmark for Privacy-Utility Trade-offs in LLM Agents — *2026*
Qiaoyuan Zheng, Yiqu Yang, Qi Gao, Imanol Schlag
NeurIPS 2026 E&D Track

- Introduced a diagnostic benchmark covering 7,852 samples across 10 domains to evaluate how LLM agents balance task utility with user-defined privacy constraints.
- Separated privacy-policy dimensions and adversarial strategies in a 5 × 5 evaluation design, using deterministic scoring to identify specific failure conditions.
- Identified substantial differences in privacy protection between frontier models and smaller open-weight models, with the weakest evaluated models leaking over half of protected attributes.

Are Near-Tied LLM Rankings Robust to Family-DIF-Guided Benchmark Recomposition? — *2026*
Qiaoyuan Zheng, Yiqu Yang
NeurIPS TAE Workshop, Under review

- Investigated the robustness of near-tied LLM rankings across five benchmarks using a spectral approximation to multidimensional item response theory and differential item functioning analysis.
- Used owner-disjoint evaluation folds and matched-random controls to distinguish systematic effects of benchmark composition from generic subtest variation.

What Makes Something Hard(er)? Explaining Question Difficulty in Natural Language 2026
Peng Cui (co-first author), **Qiaoyuan Zheng** (co-first author), Rudolf Debelak, Mrinmaya Sachan
preprint. Under Review

- Combined item response theory with LLM-based contrastive hypothesis generation to explain question difficulty, validating the explanations through held-out difficulty prediction and targeted question rewriting across reasoning benchmarks.

TUPLE-Bench: Evaluating and Mitigating Privacy-Utility Trade-offs in Typed Decision Models 2026
Yiqu Yang (co-first author), **Qiaoyuan Zheng** (co-first author), Yang Tian, Yu Fan, Elliott Ash
Under Review

- Introduced a benchmark of 720 synthetic counterfactual pairs across three domains to evaluate privacy leakage and downstream utility across five output-release settings.
- Showed that probability and confidence fields expose protected information beyond authorized decisions, increasing secret recovery by 11.46–23.61 percentage points over choice-only outputs while reducing downstream cost.
- Developed workflow-constrained privacy quantization (WCPQ), reducing secret recovery by 10.07–24.65 percentage points across three models while preserving all action branches and costs across 18 frozen workflow/cost combinations.

SkillEval: Learning Interpretable Ability Profiles of LLMs via Cognitive Diagnosis Models 2026
Berke Arda, Peng Cui, **Qiaoyuan ZHENG**, Rudolf Debelak, Mubashara Akhtar, Mrinmaya Sachan
Under Review

- Proposed a framework combining LLM-based skill discovery with neural cognitive diagnosis to construct interpretable ability profiles for 3,811 LLMs across 9,523 items from five benchmarks.

Last Translation Benchmark 2026
Vilém Zouhar, Niyati Bafna, Mukund Choudhary, ..., **Qiaoyuan Zheng**, ...
Preprint

- Contributed translation test cases and reviewed English, Chinese, and French benchmark submissions.

Selected Projects

Sorbonne University
Aperiodic Tiling Generation — *Mar 2024 - Jul 2024*

- Implemented Penrose, Wang, and Hat tiling algorithms in Python.
- Optimized generation through duplicate detection and spatial pruning; analyzed runtime scaling.

Python | Algorithms

ESIEE Paris
Q-Learning and SARSA in Grid Worlds — *Jul 2023 - Aug 2023*

- Implemented and compared Q-learning and SARSA across 15 custom grid-world environments.
- Built a map editor and environments adopted for master's-level teaching at UPEC.

Python | Reinforcement Learning

4EU+ Alliance Research Program
COVID-19 Mobility Analysis — *Feb 2020 - Dec 2021*

- Led a four-person team studying cross-country mobility patterns during COVID-19.
- Used linear regression and visualization to examine associations between public-health policies and time spent at home.

Python | Statistical Analysis

Skills

Languages ●●●●○ Chinese (Native) | English (Fluent, IELTS 7.5, GRE 328) | French (Fluent, Delf B2)
Programming ●●●●● Python | C | Java | R | Ocaml | SQL
Tech Stack Docker | Git
Frameworks & Lib Pandas | NumPy | Scikit-learn | Pytorch
Personal Interests Musical instruments (Drums, Saxophone, etc.) | Cooking | Reading | Calligraphy